

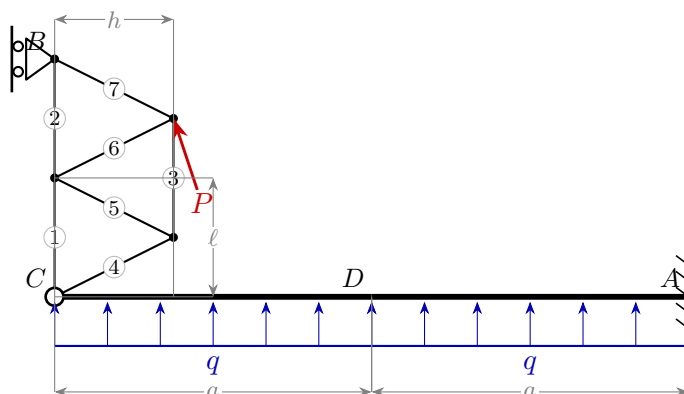
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VU Mechanik – LAWI 100515 (PI)

Exam Group A

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Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

 $A: R_x = -6.25, R_y = -47, M = 248 \quad B: R_x = 11.25, R_y = 0$
 $C: H_x = 6.25, H_y = 15$

3. Truss member forces

Member	Force S [kN]	Type
1	11.87	tension
2	5.62	tension
3	6.25	tension
4	6.99	tension
5	-6.99	compression
6	6.99	tension
7	-12.58	compression

red: tension (+) blue: compression (-)



4. Section-force diagrams of the beams

